

DC BRUSHLESS

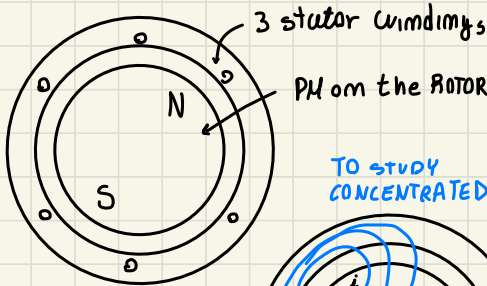
# DC BRUSHLESS NEW MACHINE!

[10/05]

used in a lot of applications... mainly for **low power/low cost**  
(for example in ROBOTICS, TOYS, DRONE.. hard disk...)



similar to **AC Brushless** (similar structure)



main difference with AC Brushless..

**stator windings are concentrated**  
(NOT distributed)

TO STUDY CONCENTRATED WINDING... consider:

and simplify the approach

consider for example (to understand the effect) a ROTOR winding (easier to study than PM) ↓ coil with current  
(flux line produced by coil on ROTOR)

$$\oint \vec{H} \cdot d\vec{e} = Ni = \underbrace{i_{\text{ROTOR}}}_{\text{total magnetomotive force}} \text{ with } N=1$$

$N := \# \text{ turns}$

Computing...

by definition of FLUX LINE!

times where **B is constant along the line**

current inside cross section

(with B tangent to the line..)

choosing one line... the line integral is the total current cross-sectioned by this line area

⇒ amp relationship  $B = \mu H$  ( $\mu := \text{permeability}$ )

flux density      magnetic flux

(\*) on that line, being B constant → you have two values of H

dividing the line..

|      |                         |                                                  |
|------|-------------------------|--------------------------------------------------|
| air: | $H_0 = B / \mu_0$       | field inside air (along $\delta$ )               |
| fe:  | $H_{fe} = B / \mu_{fe}$ | field inside ferromagnetic material (on fe path) |

$$Ni = \oint \vec{H} \cdot d\vec{e} = \underbrace{\int_{\text{length fe}} \frac{\vec{B}}{\mu_{fe}} \cdot d\vec{e}}_{\substack{\mu_{fe} \gg \mu_0 \\ \text{negligible!}}} + \underbrace{\int_{\text{length air}} \frac{\vec{B}}{\mu_0} \cdot d\vec{e}}_{\text{(in air gap)}}$$

(\*) With  $\mu_{fe} \sim 8000 \mu_0 \Rightarrow H_{fe} \approx 0$  neglected! ( $H_{fe} \ll H_0$ )  
neglect fe contribution... like "short circuit" for the flux

$\delta := \text{air gap}$

easy to compute!  $\vec{B} \cdot d\vec{e} = B \delta e$

PROPERTY: line path flux are orthogonal to surface, radial

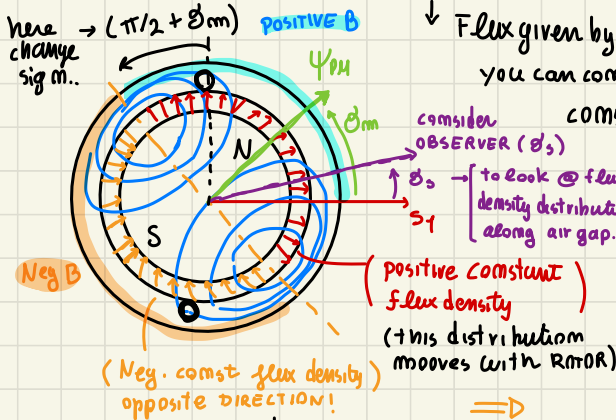
OVERALL..  $\oint \vec{H} \cdot d\vec{e} = \frac{B}{\mu_0} \cdot 2\delta = Ni \Rightarrow B = \frac{Ni \mu_0}{2\delta}$

$B = \frac{N \cdot i \cdot \mu_0}{2g} \rightarrow$  same B inside air gap.. (flux density constant in the air gap!)

↓  
all lines has same flux density (fe as a short circuit)  
(related to AIR GAP S)

↓ NOW (CONSIDER REAL machine structure)

considering a PM on the machine... PM has some effect of a coil



(looking to Flux density on  $S_1$ )

$\theta_s = 0$ . max flux density positive until  $\pi/2 \rightarrow$  then decrease... after  $\pi/2 + \delta_m$

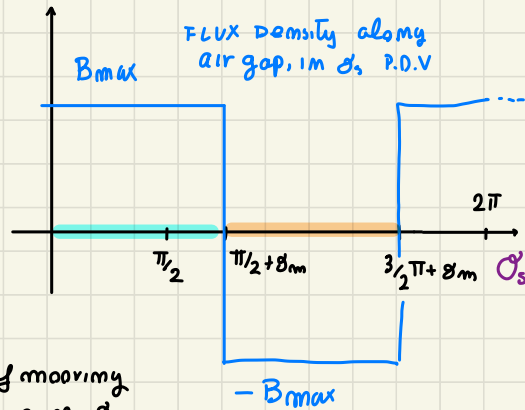
then neg. until  $3/2\pi + \delta_m$  (where change sign again)

[for  $\delta_m = 0$ , square waves with  $\pi/2$  change of sign!] ← density distribution changes (move) the B ( $\theta_s$ ) shift with  $\delta_m$  value of ROTOR POSITION

↓ Flux given by PM similar to coil flux.

you can consider stator..

consider only 1<sup>st</sup> stator coil..



so, if moving the ROTOR,  $\delta_m$  change, so this flux

(if all along S)  
all surface to find flux!



respect  $\psi_{s1}$  given by PH. (when  $i_{s1}, i_{s2}, i_{s3} = 0$ )

I need to find all the line through that surface to find dip and then integrate

$\psi_{s1} = \dots$   
 $i_{s1} = 0$   
 $i_{s2} = 0$   
 $i_{s3} = 0$

EFFECT of only ROTOR PH

← flux linked with this  $S_1$  coil of stator

is the real flux  $\phi$  (magn. flux) \* N (# turns)

$\psi = N\phi$  (flux linkage)

with  $\phi = \iint \vec{B} \cdot \vec{m} \, dS$

↑ SURFACE integral

↑ ORTOGONAL to SURFACE VECTORS

to compute.. define dS

to define dS to calculate  $\phi$ .

(if closed surface same in/out,  $\phi = 0$  total)

define  $e :=$  height of the stator,  $r_{avg} :=$  average value of radius on air gap  $\Rightarrow$

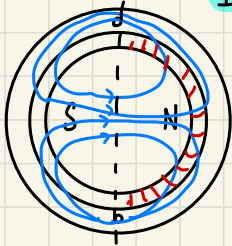
$$ds = r_{avg} \cdot \ell \cdot d\theta$$

↓

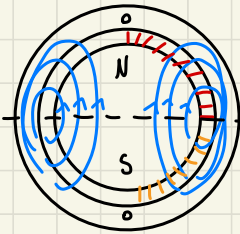
$\ell := \text{STATOR length}$

to evaluate  $\oint$  along the air surface gap..

In this 2 conditions .. What about flux linkage due to PH?



all positive! this surface give max value!  
flux lines



(no line linked...)

Q flux linkage on this case!

positive path ( $>0$ ) + B } overall Q contribute  
neg path ( $<0$ ) - B

[ While in the middle condition .... similar reasoning ]

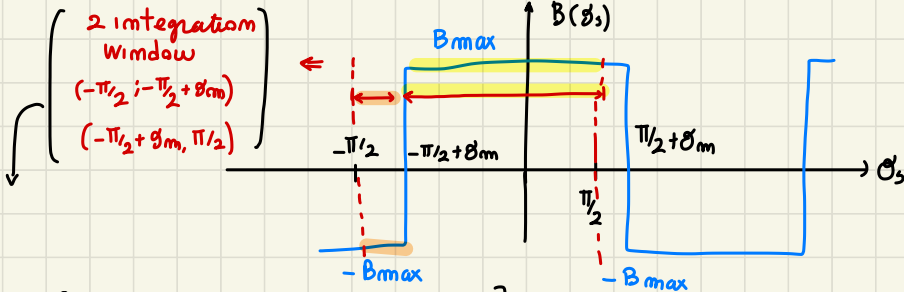
→ so overall

$$\Psi_{s1} \Big|_{i_{s2,3}=0} = N \oint \vec{B} \cdot d\vec{s} = N \int_{-\pi/2}^{\pi/2} B(\theta_s) \cdot r_{avg} \ell d\theta_s$$

$\vec{B} \times \vec{n} = B$  (due to  $B \perp m$ )

surface integral  $\Psi = \oint \vec{B} \cdot d\vec{s}$

$B(\theta_s) = \begin{cases} +B_{max} \\ -B_{max} \end{cases}$



$$\Psi_{s1} \Big|_{i_{s2,3}=0} = N r_{avg} \ell \left[ \int_{-\pi/2}^{-\pi/2 + \theta_m} -B_{max} d\theta_s + \int_{-\pi/2 + \theta_m}^{\pi/2} B_{max} d\theta_s \right] =$$

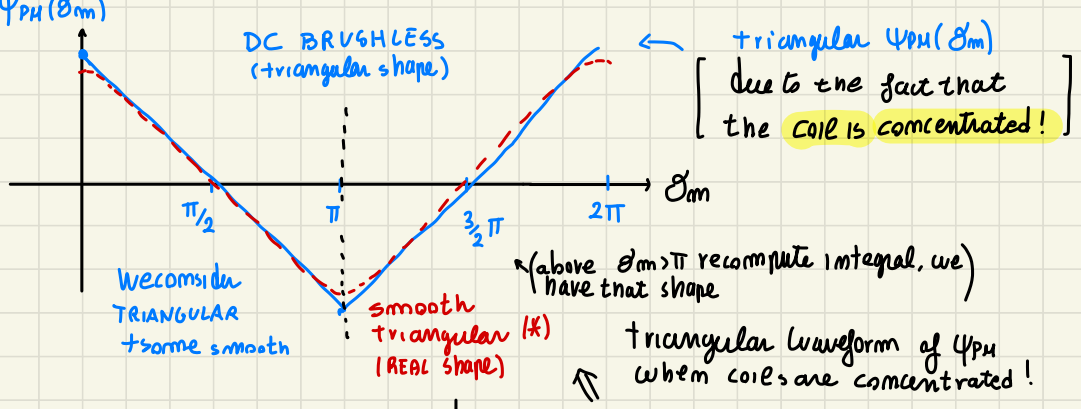


$$\Psi_{S1} = N \cdot r_{avg} \cdot l \left[ -B_{max} \left( -\pi/2 + \theta_m + \pi/2 \right) + B_{max} \left( \pi/2 + \pi/2 - \theta_m \right) \right] =$$

$$N r_{avg} l B_{max} (\pi - 2\theta_m) = \Psi_{PM}(\theta_m)$$

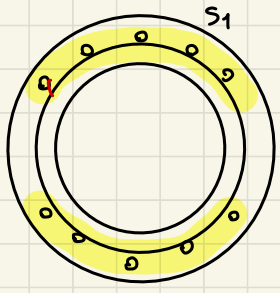
flux of PM linked with  $S_1$  when  $\alpha_{is} = 0$  as function of  $\theta_m$  (mech angle)

$\Psi_{PH}(\theta_m)$   
 (max for  $\theta_m = 0$ ) (0 for  $\theta_m = \pi/2$ ) (min for  $\theta_m = \pi$ )



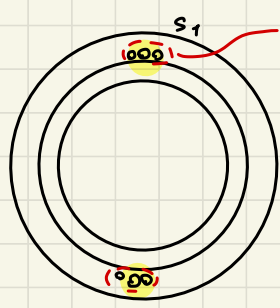
triangular waveform of  $\Psi_{PM}$  when coils are concentrated!

While on AC Brushless



distributed  $S_1$  winding

DC Brushless



concentrated  $S_1$  winding

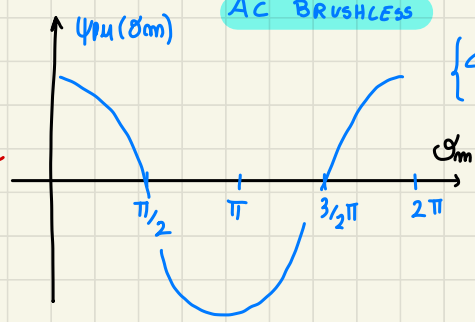
concentrated coils so you have an (\*) effect of small distribution,  $\rightarrow \Psi_{PM}(\theta_m)$  has effect of smooth for this (not a single point!) more coils inside

main difference between AC/DC brushless

AC BRUSHLESS

there is a displacement of all the coils flux linkage... so you sum the effect and obtain...

its like sum the effect of displaced triangular waveform of concentrated windings  $\Rightarrow$



cosinusoidal waveform of the flux linkage given by PM (sometimes properly shaping PM)

as done for AC Brushless... (same conceptual Reasoning...)

## DYNAMIC Equation DC Brushless

$$U_{s1} = R_s i_{s1} + p \psi_{s1}$$

$$U_{s2} = R_s i_{s2} + p \psi_{s2}$$

$$U_{s3} = R_s i_{s3} + p \psi_{s3}$$

← usual coil equation

Anisotropy as disturbance on the machine!

↑  
(anisotropic, introduce oscillation on machine)

(IF anisotropic, inductance depends on  $\theta_m$ )

$$\psi_{s1} = \overset{\text{SELF}}{L_{ss}(\theta_m)} i_{s1} + \overset{\text{II-I MUTUAL}}{M_{ss}(\theta_m)} i_{s2} + \overset{\text{III-I MUTUAL}}{M_{ss}(\theta_m - \frac{2}{3}\pi)} i_{s3} + \psi_{PM}(\theta_m)$$

same as AC Brushless But different  $\psi_{PM}$ !  
 ≠ from AC brushless

- ISOTROPIC structure hypothesis!
- connection of coil (star/delta) such  $i_{s1} + i_{s2} + i_{s3} = 0$

simplify ↓

$$U_{s1} = L_s i_{s1} + \psi_{PM}(\theta_m)$$

With  $L_s = L_{ss} - M_{ss}$

synchronous inductance

↓ [ In AC Brushless we used space-phaser, because  $\psi_{PM}$  was  $\cos(\theta_m)$ , so you find a constant phasor  $\bar{\psi}_{PM}$  → while now triangular hard to deal with! ]

↑ space phasor HARD to apply on that waveform

so in DC Brushless is better if you work with Phase Quantities

(different theory than AC brushless: don't use Clark-Park...)

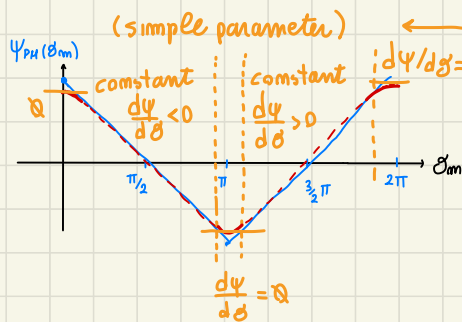
↓ dyn equations

$$\begin{cases} U_{s1} = R_s i_{s1} + L_s p i_{s1} + \frac{d\psi_{PM}(\theta_m)}{dt} e_{s1} \\ U_{s2} = R_s i_{s2} + L_s p i_{s2} + \frac{d\psi_{PM}(\theta_m - \frac{2}{3}\pi)}{dt} e_{s2} \\ U_{s3} = R_s i_{s3} + L_s p i_{s3} + \frac{d\psi_{PM}(\theta_m + \frac{2}{3}\pi)}{dt} e_{s3} \end{cases}$$

ISOTROPIC structure  
 "d flux" ( $L_s$  instead of  $L_{ss} - M_{ss}$ )  
 $\frac{d}{dt}$  what is this?  
 (I call it emf as usual!)

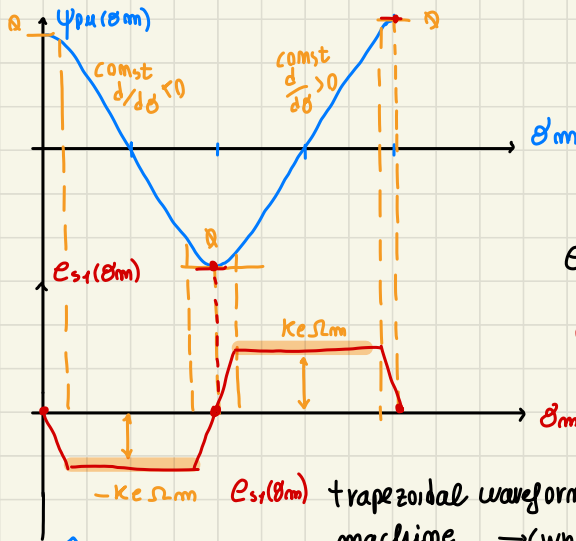
$$\frac{d\psi_{PM}}{dt} = \frac{d\psi_{PM}(\theta_m)}{d\theta_m} \cdot \frac{d\theta_m}{dt} =$$

$$= \frac{d\psi_{PM}(\theta_m)}{d\theta_m} n_p \Omega_m = e_{s1}$$



ELECTROMOTIVE FORCE  $e_s$  depending on the speed  $\Omega_m$  and  $d\psi/d\theta$  (parameter)

→ Let's plot  $e_s(\theta_m) \Rightarrow$



$K_e = m_p \frac{d\psi_H}{d\theta_m}$  when it is constant

$$e_{s1}(\theta_m) = \left[ \frac{d\psi_H(\theta_m)}{d\theta_m} \right] m_p \Omega_m$$

(due to coil not point shape... you need a smoothness...)

so this is NOT RECTANGULAR because of this "smooth" effect!

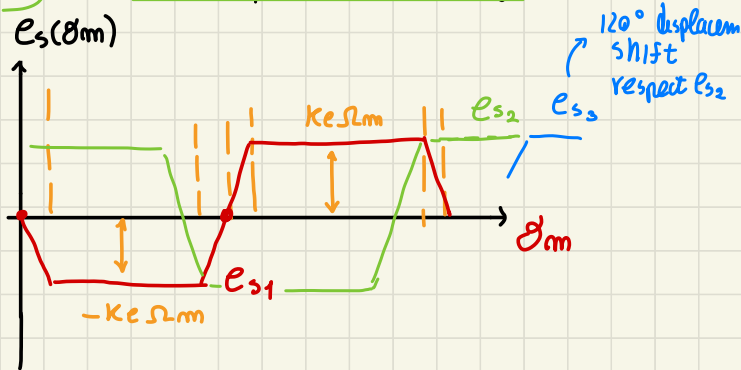
while trapezoidal

↑ MACHINE called "DC BRUSHLESS Trapezoidal emf"

trapezoidal waveform of emf  $e_s$  in DC Brushless machine... → (while in AC Brushless was sinusoidal)  $i_c(t)$  was sinusoidal,  $\psi = \cos(\theta_m) \rightarrow \frac{d\psi}{dt} \sim \sin(\theta_m)$

respect the other emf.. (om s2, s3)  
 $e_{s2}$  has same shape with 120° shift

emf each other with 120° displacement!



rewriting:

Dyn equation

$$\begin{cases} V_{s1} = R_s i_{s1} + L_s \frac{di_{s1}}{dt} + e_{s1} \\ V_{s2} = R_s i_{s2} + L_s \frac{di_{s2}}{dt} + e_{s2} \\ V_{s3} = R_s i_{s3} + L_s \frac{di_{s3}}{dt} + e_{s3} \end{cases}$$

⇒ ENERGY BALANCE (looking 1<sup>st</sup> phase)

$$V_{s1} i_{s1} = \underbrace{R_s i_{s1}^2}_{\text{Joule losses on } R_s} + \underbrace{L_s i_{s1} \frac{di_{s1}}{dt}}_{\text{variation of energy stored into inductance}} + \underbrace{e_{s1} i_{s1}}_{\text{Pm mech power}}$$

do the same for 2<sup>nd</sup> and 3<sup>rd</sup> coil

⇒ OVERALL ⇒ ...

(constant inductance  $L_s$  make easier analysis)  
 $\frac{dU}{dt}$  ( $U = \frac{1}{2} L_s i_{s1}^2$ )

TOTAL  $P_m$  given by  
contribute of the 3 coils

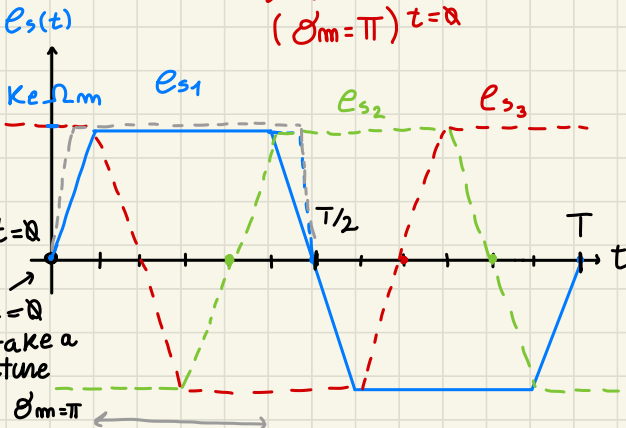
$$P_m = e_{s1} i_{s1} + e_{s2} i_{s2} + e_{s3} i_{s3}$$

to understand  
what happens:

↓ representing  $e_s(t)$  as function of time: being  $e(\theta_m)$  trapezoidal

START from  
( $\theta_m = \pi$ )  $t=0$

... With constant speed  $\Omega_m$ ,  
 $e(\theta_m)$  remain const over time...  
same shape in time



↙ @  $t=0$  when start the ROTOR  
initially is ↓

(consider ccw motion,  $\Omega_m > 0$ )

IF I check for  $\theta_m = 0$  initially...

with  $\Delta t$  also  $\Delta \theta$  and  $\Omega_m > 0$

$\Delta t = \frac{\Delta \theta}{\Omega_m} \rightarrow$  goes negative! meaningless

(I start with  $e_{s1}$ )

↓ while for  $\theta_m = \pi$  as we see  
from  $e(\pi)$ , we have  $e_{s1} \uparrow$   
increasing

↙ I take a picture for  
( $\theta_m = \pi$ )

maintain this pick length  
~  $120^\circ$  approx, even  
IF NOT TRUE... period limited  
by subperiod → SIMPLIFICATION...  
EASY TO DRAW

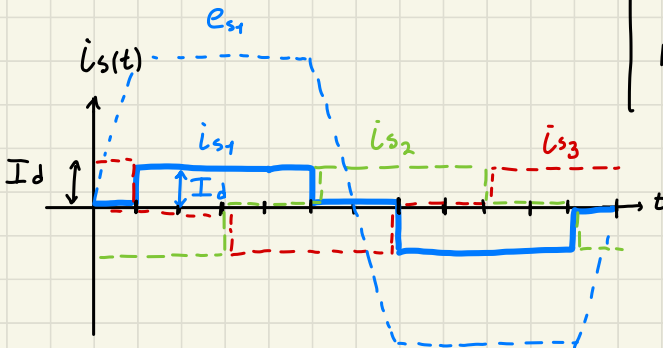
maintain  $\geq 120^\circ$  ~ square wave ( $e(\theta_m) \sim$  rectangular) usually  $e_{s1}$  is constant  
for a period higher than  $120^\circ$   
(trapezoidal  $e_{si}$  with  $120^\circ$  shift of amplitude  $Ke \Omega_m$  as max)

⇒  $e_{s1}, e_{s2}, e_{s3}$  trapezoidal waveform (with  $120^\circ$  displacem.)

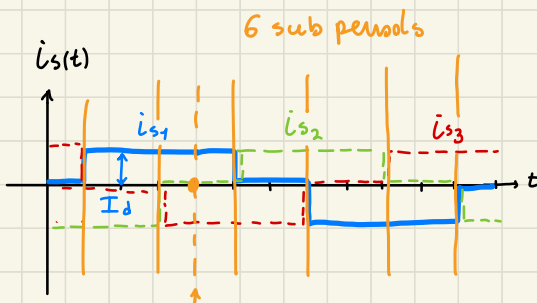
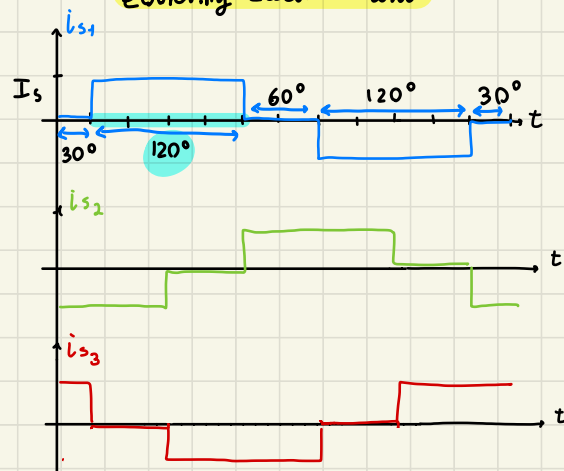
↓  
• to have Constant  $P_m$ ... current shape? such that  $i_{s1}, i_{s2}, i_{s3} : P_m = \text{const?}$

SOLUTION:

[ considering  
 $120^\circ$  of the 3 current  
as square waves ]



# Looking at each current:



@ this t:  $i_{s1} = I_d$   $i_{s2} = 0$   $i_{s3} = -I_d$

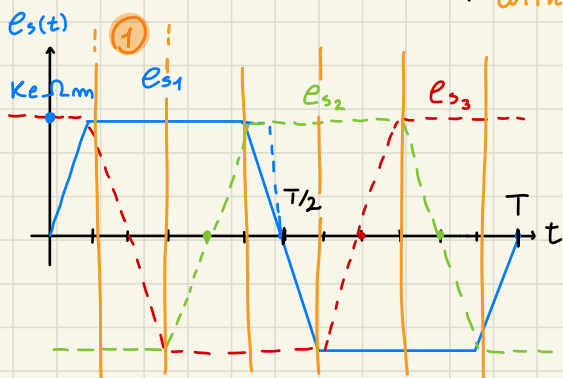
$\Rightarrow i_{s1} + i_{s2} + i_{s3} = 0$

only square waves such  $\sum i_{si} = 0$

(only square waves which guarantee  $\sum i_{si} = 0$ , with values of 120° max/min.. shift)

At you have different configuration

with 6 sub periods



We need  $e_s(t)$  constant for at least 120°!

(\*) in the time interval where  $e_{sj}(t)$  is variable,  $i_{sj} = 0$  so there is no contribution to the mech power

(otherwise you have oscillations)

TRUE IF the 1st demy which  $e_{sj}$  is variable has to be  $\leq 60^\circ$ !

(otherwise there is  $e_s$  variation in other periods!)

so as discussed before  $\Rightarrow$  you need  $\Delta t \geq 120^\circ$   
 $e = k_e \Omega_m$  constant!

$\hookrightarrow$  in each SUB-PERIOD define  $e_{s1}, i_{s1} / e_{s2}, i_{s2} / e_{s3}, i_{s3} \dots$

(FOR EXAMPLE)

6 sectors in which you have some combinations of the three currents and emf

between  $30^\circ \div 90^\circ$  (60° period)

$$\begin{cases} e_{s1} = k_e \Omega_m & i_{s1} = I_d \\ e_{s2} = -k_e \Omega_m & i_{s2} = -I_d \\ e_{s3} \sim \text{variable} & i_{s3} = 0 \end{cases}$$

$\rightarrow$  remove variable 3rd case give contribut.

$\leftarrow$  no contribution to  $P_m$  (\*)

period  $\leq 60^\circ$  where  $e_s$  is variable!

guarantee

$P_m \approx \text{const}$

otherwise oscillation on  $P_m$ !

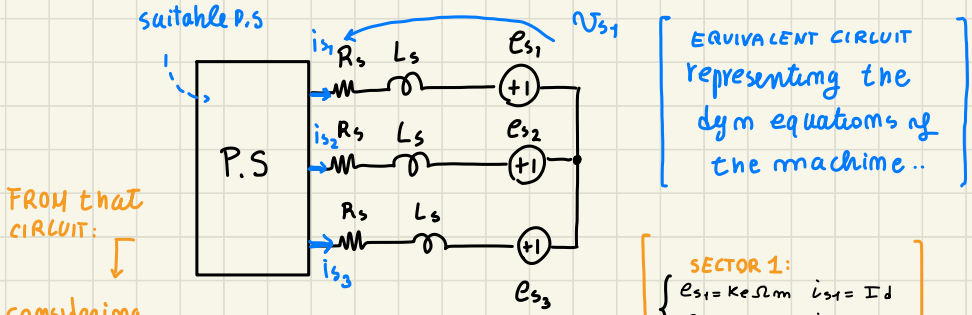
$$P_m = \underbrace{k_e \Omega_m I_d}_{e_{s1} i_{s1}} + \underbrace{(-k_e \Omega_m)(-I_d)}_{e_{s2} i_{s2}} = 2 k_e \Omega_m I_d$$

constant torque

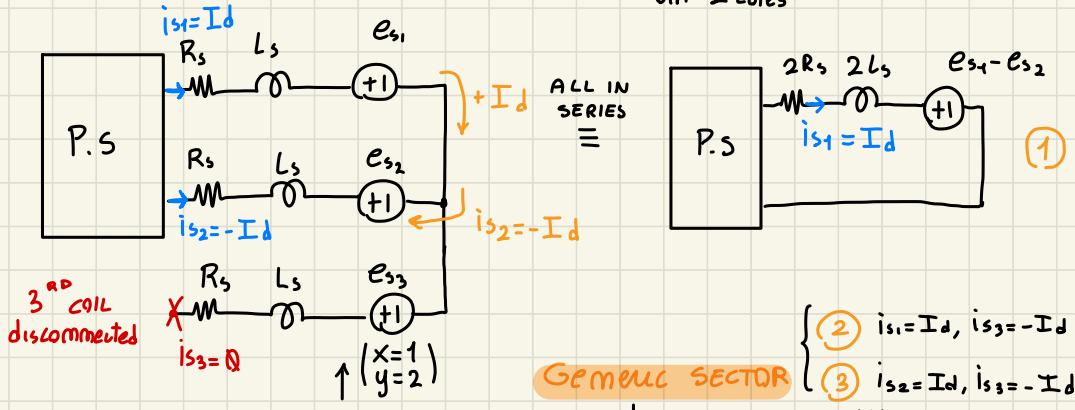
intuitive current with square wave amplitude

$\Rightarrow$  so the torque is  $T_e = P_m / \Omega_m = 2 K_e I_d$  ← in the time interval where  $e_{sj}(t)$  is variable,  $i_{sj} = 0$  so there is no contribution to the mech power  
 (simple approach!) Te depends on this amplitude

considering the **Equivalent circuit** (seen by P.S) (from P.S POV)



**SECTOR 1** during  $1/6$  of total  $T$  ( $60^\circ$ )  
 is like having  $i_{s3} = 0$  (disconnected coil 3)  $\downarrow$  P.S WORKS only on 2 coils

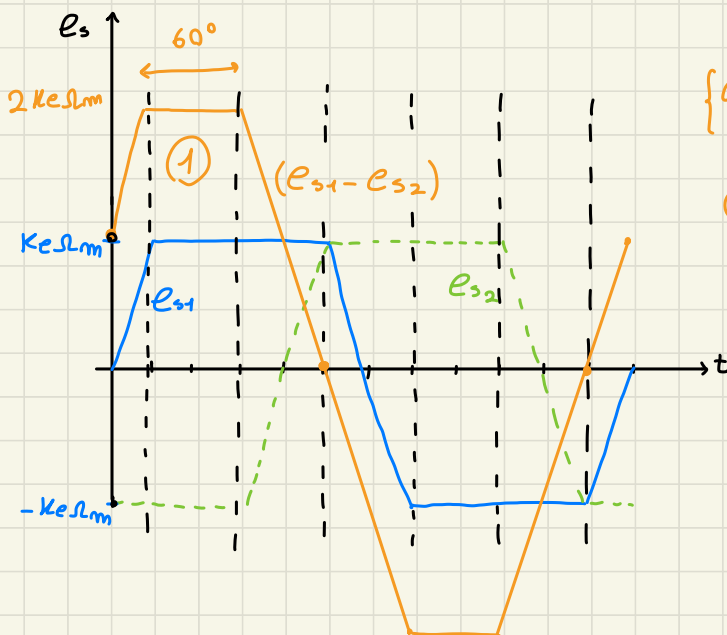


$x, y$  depends on sector  
 $\downarrow$   
 v sector 2 coils connected in series and a third one disconnected

$I_d$  generic current..  
 fictitious current, I want  $i_d = I_d$  to maintain torque constant  $\rightarrow$   
 I wanna obtain that current  $i_d$  (sometimes  $i_{s1}, i_{s2}, -i_{s3} \dots$ )  $\Rightarrow$  FICTITIOUS current maintained  $I_d$  for TORQUE CONTROL

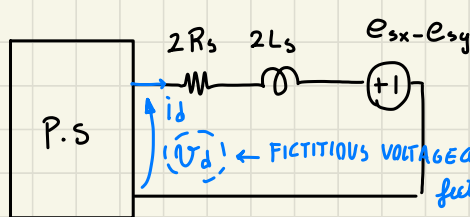
total emf  $(e_{sx} - e_{sy})$   
 ( $x, y$  depending on the sector)

LOOKING at That Plot: (to compute  $e_{sx} - e_{sy}$  effect on  $t$ )



{ amplitude  $2K_e \Omega m$  max }  
maintained for  $60^\circ$

①: time to time electromotive force (concatenated) (is doubled)  
↑  
during FIRST period ①  
on equivalent circuit  
we have a total  
emf (time to time emf)  
has amplitude  $2K_e \Omega m$ !



$$U_d = 2R_s i_d + 2L_s \frac{di_d}{dt} + \underbrace{2K_e \Omega m}_{\text{total emf of each sector } j=1, \dots, 6}$$

total emf of each sector  $j=1, \dots, 6$   
total emf on each sector

and  $(e_{s1} - e_{s2})$  is the same with  $60^\circ$  shift respect  $e_{s1} - e_{s2}$   
... a lot time to time voltage with  $e_{sx} - e_{sy} = 2K_e \Omega m$  during that sector  
↓ (TO HAVE CONSTANT POWER:)

from energy balance on (\*)  $\Rightarrow U_d i_d = \underbrace{2R_s i_d^2}_{\text{JOULE LOSSES}} + \underbrace{2L_s i_d \frac{di_d}{dt}}_{\text{VARIATION OF ENERGY STORED}} + \underbrace{2K_e \Omega m i_d}_{P_m}$   
on the equivalent circuit (equivalent on power)

$$\Downarrow \left( T_e = \frac{P_m}{\Omega_m} \right)$$

energy balance on equivalent circuit (maintain POWER)

$$T_e = 2K_e i_d$$

(same result as before!)

[constraint:  $P_m \sim \text{const}$  if  $e_{s1}, e_{s2}, e_{s3} \sim \text{const}$  for  $120^\circ$  at least]  
 ↳ NOT the true constraint (it is easy to explain but too restrictive)

To have  $P_m = \text{const.}$  (True constraint!) || We want this total emf (line to line)  
 $e_{sx} - e_{sy}$  is constant for  $60^\circ$  at least ||  
 (line to line emf constant for  $60^\circ$  at least)

because  $\downarrow$  (machine designed to have  $\geq 60^\circ$  of constant total emf)  
 { P.S used to supply line to line emf  $\rightarrow$  NOT single emf (P.S sees  $e_x - e_y$ )  
 $P_m \text{ const. is line to line emf } \sim \text{const}$  for at least  $60^\circ$  (NOT single  $e_s$ ) }

• NOW:  
 suppose to define  $\begin{cases} R_a = 2R_s \\ L_a = 2L_s \\ K_{epm} = 2K_e \end{cases}$  [parameter of (PM) DC machine] same param of old explained machine!  
 With equation

dyn eq of equivalent circuit of DC brushless  $\equiv$  eq circuit of PM DC mach.  
 (you have same eq. circuit)  $\rightarrow \begin{cases} U_a = R_a i_a + L_a \dot{i}_a + E \\ E = K_{epm} \Omega_m \end{cases}$  } PM DC machine equation  
 III  
 DC brushless!

Same idea, to control the torque I must maintain  $I_d \sim \text{const}$  (same  $T_e$  control idea of PM DC mach)

$\Rightarrow$  Control P.O.V  $\rightarrow$  similar to armature dyn eq.  
 + I need control constant current ( $i_d = I_d$ ) (fictitious current) CONTROL P.O.V (like control of  $i_a$  on DC PM)

$\downarrow$   
 "DC" Brushless because works like DC machine

BUT it is NOT DC (3 phase machine)

Looking at the current:



NOT DC  $\rightarrow$  Alternating machine 3 phase

but in the control P.O.V  
 In each sector the behaviour of control is the same of PM DC machine

due to how P.S work on that mach:  
 (there is NOT a commutator, BUT the

P.S electronically commutate the phase ... using sometimes  $i_{s1}, i_{s2}$  OR  $i_{s1}, i_{s3}, \dots$ )

$\leftarrow$  ("electronically commutated machine")  
 other name of DC Brushless



15/05 → PROF. Ferdinando Mapelli



POWER ELECTRONICS, manufacturing of electrical machine for  
Low Voltage - high power application (Agriculture)

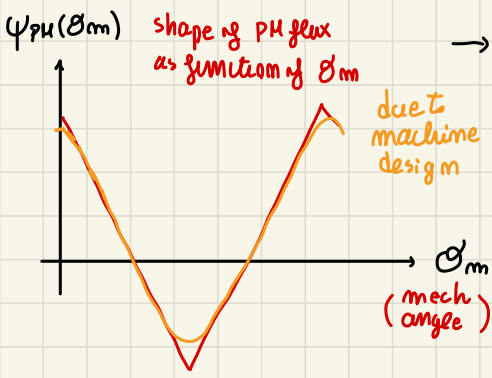
22/05 → J. Fran... electrical Drives for Industrial application

[DC Brushless problems, Switch Reluctance machine] next...

↳ face the DC Brushless problem  
and introduce SWITCH RELUCTANCE MACHINE →

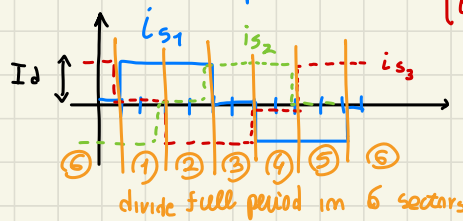
...DC BRUSHLESS [CONTINUE...] simple machine ... low cost / low power machine a lot used

MAIN characteristic:



use phase quantities (non sinusoidal, trapezoidal emf...)

IDEA: to realize current with this shape:



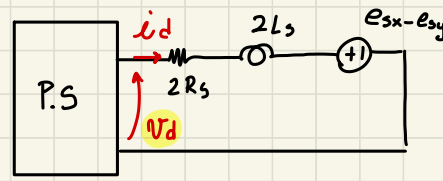
so we don't use space phasor!

(control idea!)

have this current form

Idea to control the Torque given by:  
 $T_e = 2k_e I_d$

From the equivalent circuit seen by P.S POV:



for each sector: one current remain  $i_z = 0$  while  $i_x = -i_y = I_d$ : equal opposite

to have

proper control: strategy to have  $T_e$  constant

- constant torque you need to have:
- $i_d = \text{const} = I_d$  (reference)
- $E_x - E_y$  constant for the sector period at least  $60^\circ$
- $[2k_e \Omega_m] \leftarrow \text{constant value... for at least } 60^\circ \text{ of time!}$
- (line to line emf)

from Pym eq.

$$V_d = 2R_s i_d + 2L_s \frac{di_d}{dt} + 2k_e \Omega_m \dot{\theta} \approx \text{similar to PM DC machine (from equivalent circuit)}$$

$T_e = 2k_e i_d$   
usual control scheme } control scheme  $T_e$  depends only on fictitious current  $i_d$  with only one time constant!

→

Dyn eq:

quantity which modify  $i_d$ !

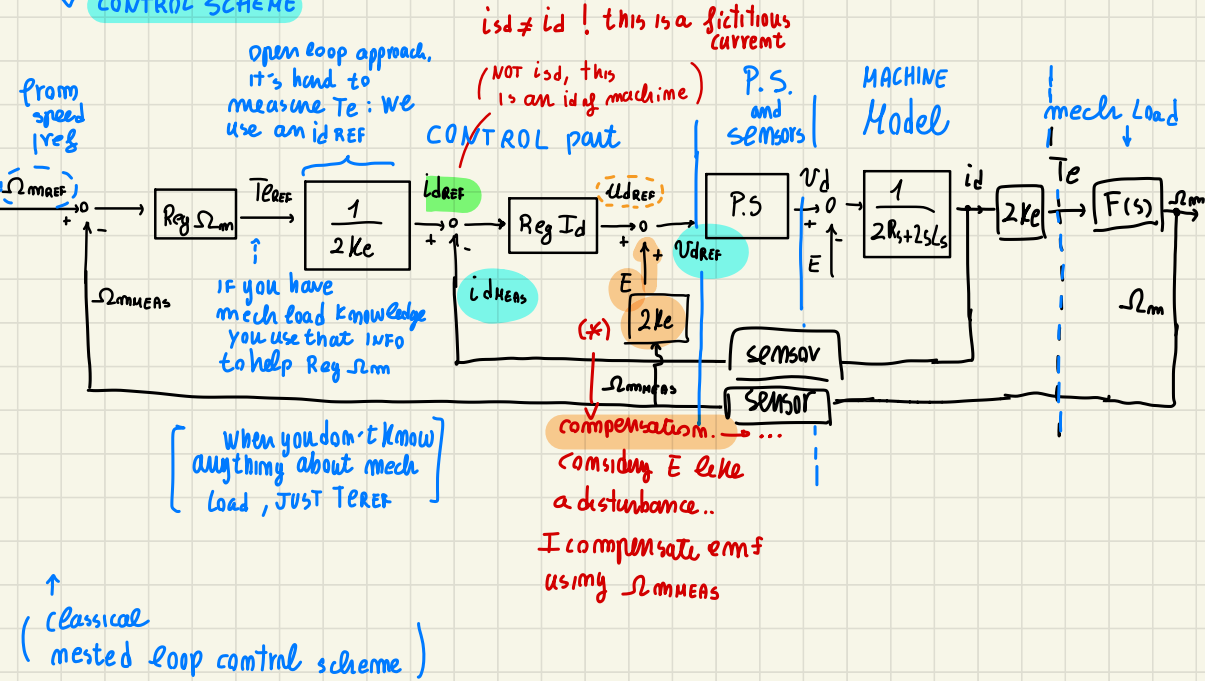
FICTITIOUS equation of the machine  
(the real machine is 3 phase)

$$V_d = 2R_s i_d + 2L_s \frac{di_d}{dt} + 2k_e \Omega_m$$

it can be  $i_{s2}, i_{s2}, -i_{s3}, -i_{s2} \dots$

$$T_e = 2k_e i_d \leftarrow \text{torque depends only on } i_d \text{ (fictitious current)}$$

## CONTROL SCHEME



## (\*) Without the E compensation

output of (RegId) is send directly on P.s, total voltage request...

you still reach TREF: Because the error increase and we increase overall request of VOLTAGE!  $\Rightarrow$  machine sometimes used in electric bike. Problem arise if you

- switch off/on while moving  $\Omega_m \neq 0 \rightarrow$  emf appear  $\neq 0$ : current control PI, start from 0 integral part  $\Rightarrow$  output of PI not enough!  
problem initially! negative forcing terms... supply and more...

$\hookrightarrow$  torque required is still reached in time  $\rightarrow$  you have no problems! When you

- When you don't compensate, if you have knowledge **don't need to start on the fly, START only with  $\Omega_m = 0 \rightarrow$  OK!**

on E oscillation you can compensate it!

$\downarrow$  When design controller of a system, if you have disturbance acting  $\rightarrow$  you can compensate by Feed Forward.

[ design a ROBUST controller on the disturbance if you have info about it.. ]

[ OR design controller NOT on T.F but on the disturbance acting ]  $\rightarrow$  more ROBUST

[ When you know something about  $d(t)$ , compensate disturb. ]

- OUR idea is to design controller only on T.F (easiest best solution) **without compensation**

- only torque / speed control on DC Brushless,

( **No need to control Flux!** )

$\rightarrow$  also in AC Brushless we had a  $\psi$  control by means of isd demagnetizing



While looking also

on our dym model we don't know how to reduce the FLUX!

(as on PM DC machine  $\rightarrow$  on which we control speed/torque, NOT flux given by PM)

FROM

control scheme: I need

**Id meas**: Id is fictitious: I need to introduce something to go from real currents  $i_{s1,2,3}$  to fictitious quantity  $i_{dmeas} / i_{dREF}$

We need a technique to



move from real three phase machine to that fictitious machine

$\hookrightarrow$  ...

**Vdref**: Reference of Vd fictitious generated P.s, BUT the power supply need some way to supply 3 terminals (NOT 2)

$i_{dREF}$  is fictitious  $\rightarrow$  I need a way to find it and supply 3 terminals from it

$\rightarrow$  how to go from real  $i_{s1}, i_{s2}, i_{s3}$  to fictitious  $i_d$ ?  
and also for  $v_d$

(similarly to what we have done on previous machines from  $dB, dq$  ref)

(NOW we are NOT using space phasor!)

$\hookrightarrow$  Now we are

using electrical quantities..  $\rightarrow$  IDEA:

from measurement of  
Real currents 3 phase  $i_{s1}, i_{s2}, i_{s3}$ ...

I need here a system that takes as  
output of the block  $i_{sd} = i_{s1}$  OR  $i_{s2}$  OR  $i_{s3}$   
OR  $-i_{s1}$  ...

$\downarrow$  choosing one of  
the three input by a "selector"

I need to choose one of  
the input by a MUX (MULTIPLEXER)

$\downarrow$  and to realize the selection  
of the  $i_{sj}$  I need to

know where is the ROTOR  
looking at  $\theta_m$   $\rightarrow$  to identify  
the SECTOR  
(not exact  $\theta_m$ )

$\uparrow$  to choose between the  
input, and realize correct selection..

I need information about  
 $\theta_m$  Where is the ROTOR?

$\downarrow$  this tell me in which sector I'm now!

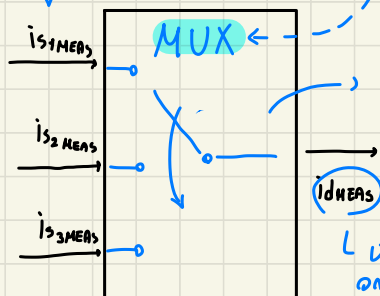
$\leftarrow$  (once we know  $\theta_m$ , each SECTOR is  $60^\circ$   
and I can identify electrical world position)

knowing period of  
Flux as function of  
 $\theta_m$ : mech angle

on elect WORLD  $\Rightarrow$  so as **Input** I don't need exact info

about  $\theta_m$ !  $\rightarrow$  instead I need to know current SECTOR

$\hookrightarrow$  usual solution to reduce the cost?  $\Rightarrow$



{sector 1:6}

$\hookrightarrow$  ... how?

so to reduce the cost to identify  $\theta_m$ ... (NOT so accurate  $\theta_m$ , BUT the SECTOR)



- I use 3 hall-effect sensors!

use the ROTOR itself (which is a PH)  
(OR sometimes a magnet on MOTOR shaft)

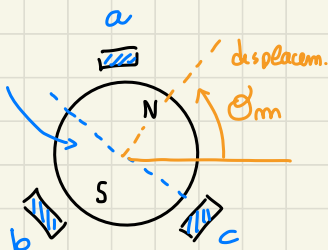
(3 hall effect sensor with  $120^\circ$  displacement)

place hall effect sensors (very cheap) (a, b, c)

[Digital sensor] → LOW COST!

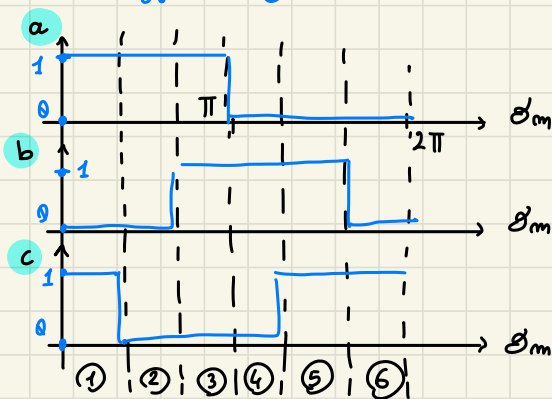
identify 1/Q if the sensor (a, b OR c) is placed  
on N/S side of ROTOR

- ROTOR OR
- some magnet on shaft with same number of mp of ROTOR



SENSOR receive 1 IF it is in front of N magnetic pole  
0 IF it is in front of S pole

hall effect signals received:



that's a square wave function of mech angle

6 combinations of sector a different comb. to identify ROTOR sector!

SIMPLE APPROACH!



for  $\theta_m = Q$  N is between a and c and so on...

6 combinations! (6 DOF) and this  
abc combination identify the SECTOR

We can use a TABLE which identify sector where ROTOR is!

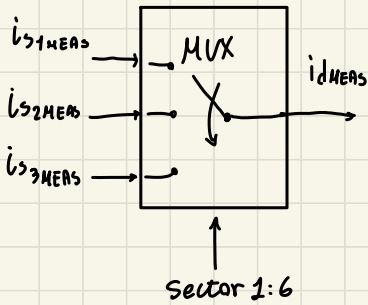
- sometimes HALL SENSOR fixed (when placed on Floppy disk stay inside core, on CDROM this core on PCB close to ROTOR)

- You should check the alignment... in ROBOT application they are placed on the shaft → so you need to TEST, move PH and understand that the sensor switch signal is aligned with emf!

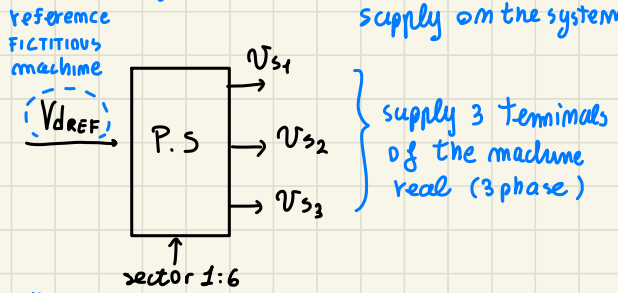
- better NOT to use ROTOR Flux, which has a field depending on producing disturbance → so you use a different sensor on shaft (something separate from machine)  
PH ROTOR magnetic flux has a field inside air gap which depends also on its (stator current) producing disturbance

no influence by is

(final block for our purpose:)



→ you also need to know how to go from  $d_{REF}$  to the real three phase supply on the system

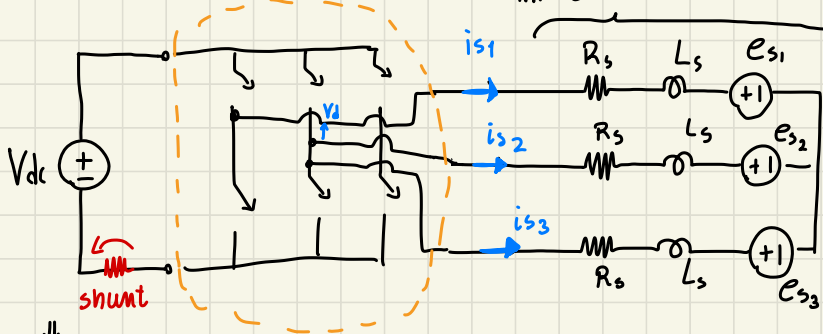


(you need to have suitable P.S to realize that  $V_{s_j}$ )

IDEA: control a constant current for each sector. for  $60^\circ$  maintain same current:  $i_{s1}$ , then  $i_{s2}$ , then  $i_{s3}$ ... constant!

you need something HERE in order to drive the machine as we want...

LOTS of combinations ↓ Realize it from a  $V_{dc}$  with classical P.S with 3 TERMINALS



↓ FROM  $V_{dc}$  := we wanna produce  $V_{dc}$  for each sector: ⇒ with DC/DC converter: control constant current (\*) ⇒ ...

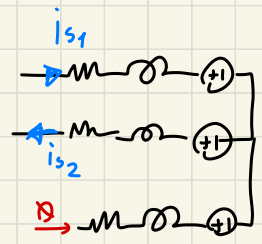
{ 4 quadrant DC/DC converter ← you need to produce voltage  $\geq 0$  with current  $\geq 0$  } 4 quadrant

you have 6 combinations using 3 legs... (NOT an inverter!)

3 legs working as 6 DC/DC converter -- with leg choice depending on sector!

=> (\*) DC/DC converter : control a constant current from  $V_{dc}$  to a constant current

↓  
SECTOR 1: control  $i_{s1} = I_d$ ,  $i_{s2} = -I_d$



↓  
so you need 2 LEGS! (NOT 1 LEG)

{ producing positive/negative voltage  
with positive/negative current

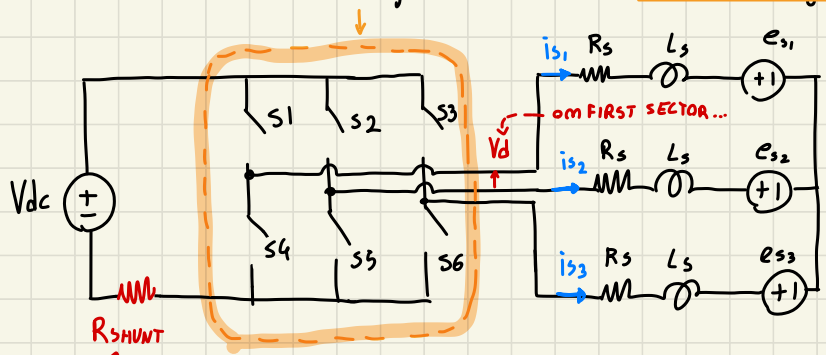
$L$ ,  $\begin{cases} v \geq 0 \text{ move exit/enter} \\ i \geq 0 \text{ Generator/motor} \end{cases} \rightarrow 4 \text{ quadrant!}$

↓  
4 QUADRANT  
DC/DC  
CONVERTER

• 2 LEGS for each sector

realized... at the end  $2 \times 6 = 12 \text{ LEGS}$

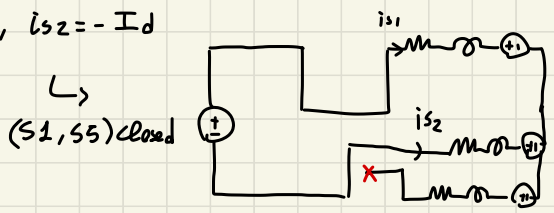
BUT IF you look to the usual configuration



3 LEGS configuration... even if it seems an INVERTER  
(IT IS NOT!) → this represent 6 DC/DC converter

so choose properly which switch to use & SECTOR

for example : SECTOR 1 we use (s1, s5) to control current  
such that  $i_{s1} = I_d$ ,  $i_{s2} = -I_d$





Use  $S1, S5$  to control current properly,

OR IM

SECTOR ②: Use  $(S1, S6)$  such  $i_{s2} = I_d, i_{s3} = -I_d \dots$

③:  $(S5, S6)$

④:  $(S1, S2)$  such  $i_{s2} = -I_d, i_{s3} = +I_d$

opposite direction of current  
respect previously ①

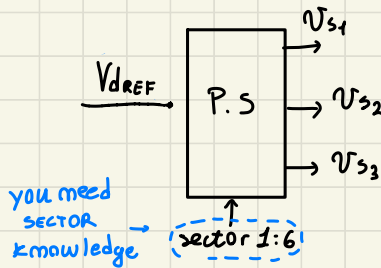
I control  
2 legs @ a time  
for each sector.

6 combinations of DC/DC converter  
using only 3 LEGS  $\rightarrow$  you use  
2 at a time

NOT AN INVERTER!  $\rightarrow$  3 LEGS working as 6 DC/DC  
converter

and the choice of the LEGS depends on  
the current SECTOR  $\{1, 6\}$

SO OVERALL to realize  $V_{s1}, V_{s2}, V_{s3}$ :



$\hookrightarrow$  sector gives you info about the  
2 LEGS to use... and  $V_{dREF}$  is

realized working by PWM technique  
to realize Voltage Reference

so switch ON/OFF  
properly to  
maintain  $V_{dREF}$   
on the equivalent  
circuit

you

work on the currently used 2 LEGS in order to realize desired  $V_d$   
 $V_d$  applied sector by sector!

$\Rightarrow$

and, chosen the 2 LEGS in use: the 3<sup>rd</sup> LEG not used on current sector  
has to stay OPEN  $\rightarrow$  IF you need to use your inverter like this type of P.S.

(usually on P.S. the two switches of each leg

$\Rightarrow$  BUT in my case IF NOT used LEG,  
Both switches of LEG OPEN!  $\rightarrow$  I need  
to be able to OPEN both! (NOT close both!))



choose properly your P.S

• you have another advantage... placing a small Resistance, called  
"shunt" resistance on the circuit (\*\*)

Looking on VOLTAGE DROP on it, you can measure the current!

that is circulating there...



which is representative of the current  
of the load OR the negative load  
current!

measure current inside the load in a very  
simple way! JUST knowing the SWITCHES  
state to know the sign of it



Very low cost solution (while in AC Brushless we needed  $i_{s1}, i_{s2}, i_{s3}...$ )

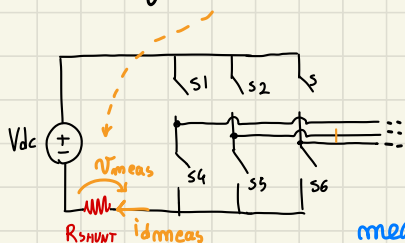
$\downarrow$  REDUCE the cost of this application

[only one "sensor" to produce information about  $V_{dREF}$ !]

+ INFO about STATE of the system and switches / sector

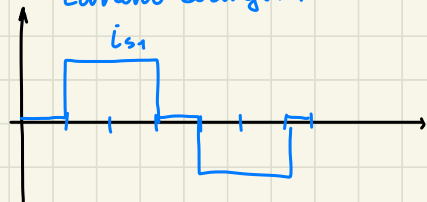
to produce INFO about  $i_{dmeas}$

$\hookrightarrow$  What about the waveform  
of  $i_{s1}$  for this type of machine? ...  
(as we discussed before, it's a square wave)



↘ In this type of machine (DC brushless)

current waveform



← it's an Alternate current machine (AC),  $i_s(t)$  is AC, with  $i_{s\text{avg}} = 0$  and alternate (3phase)

BUT for CONTROL POV you can consider it as a DC machine (that's why "DC BRUSHLESS" or

"ELECTRONICALLY COMMUTATED DC machine")

(NOT by brushes and commutator, BUT by Power electronics)

← commutate by means of the power supply!

$$\Omega_m > \Omega_b$$

- NOW Let's discuss a new aspect of this machine :

from  $V_d = 2R_s i_d + 2L_s p i_d + 2K_e \Omega_m$  (dyn equation of equivalent circuit)

no flux control, BUT anyway at a certain point you reach a limit..

↳ even if we don't control the flux, from dyneq. for  $\Omega_m \uparrow$

also  $V_d \uparrow \dots$  we have limits! speed increases voltage

↓ when you reach max voltage that P.S can apply

(BASE speed) →

↳ in steady state ( $p i_d = 0$ )

$$V_{d\text{max}} = 2R_s \overset{\text{apply max current constant}}{I_{d\text{max}}} + 2K_e \Omega_b$$

$$\| V_{d\text{max}} = V_{dc} \text{ for the converter} \|$$

$$\Omega_b = \frac{V_{d\text{max}} - 2R_s I_{d\text{max}}}{2K_e} \quad (\text{BASE SPEED})$$

↳ you can increase furthermore  $\Omega_m$  until

$$\Rightarrow \Omega_o = \frac{V_{d\text{max}}}{2K_e} \quad (\text{NO LOAD SPEED}) \quad \left\{ \begin{array}{l} \text{increase } \Omega_m \\ \text{But reach a point without torque: } T_e = 0 \end{array} \right\}$$

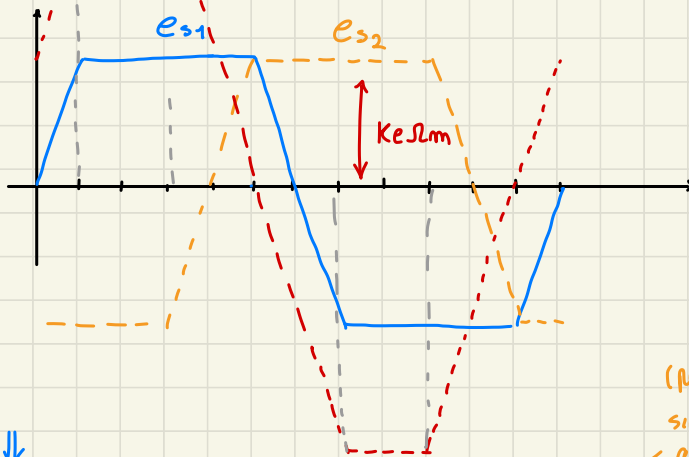
(similar behaviour to PM DC machine)

there is another way to increase the speed...  $\Rightarrow \Omega_m > \Omega_b \Rightarrow$

INCREASE the speed  $\Omega_m > \Omega_b$ : considering signal behaviour

① SECTOR  $[e_{s1} - e_{s2}] = \text{total line to line emf}$ :

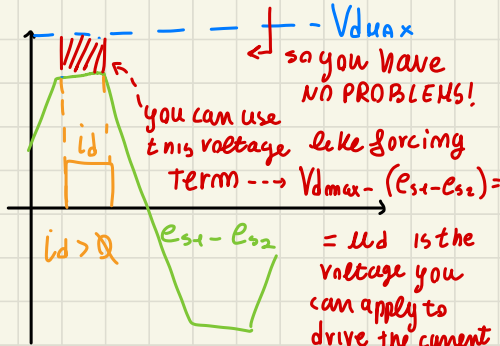
$2k_e \Omega_m$   $e_{s1} - e_{s2}$  (with amplitude  $2k_e \Omega_m$ )



(period on which  $(e_{s1} - e_{s2}) = \text{const}$ )  $\sim 60^\circ$   
 simply, consider this  $30^\circ$  point  
 as sector change

FOR  $\Omega_m < \Omega_b$

the max Voltage is higher than the total emf!

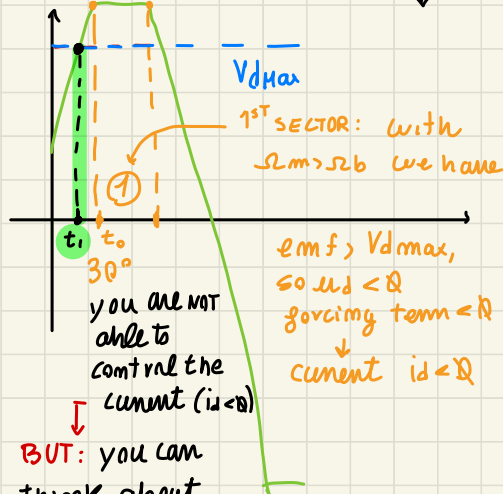


usually by PWM technique to drive current

FOR

$\Omega_m > \Omega_b$

$(e_{s1} - e_{s2}) > V_{dmax}$

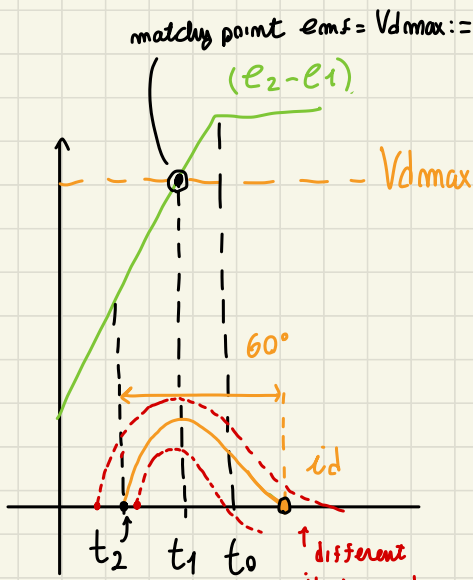


BUT: you can think about an alternative idea for control:

← by changing the sector BEFORE

NOT change sector on to BUT before, on the

point  $t_1$  where  $i_d = Q \Rightarrow$  IF I change the sector in a point  $t_2 < t_1 \dots \hookrightarrow \text{ZOOM IN} \Rightarrow$



IF change the sector in  $t_2$ ,  
I have  $V_{dmax} > e_{mf}$  (positive forcing term)

so between  $[t_2, t_1]$  the current increases... (positive forcing)

than after  $t_1$ , reached  $i_{dmax}$  in  $t_1$

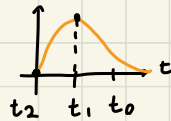
~~$V_d = 2R \cdot i_d + 2L \cdot \frac{di_d}{dt} + 2k_e \omega_m$~~  ( $V_d = 2k_e \omega_m$ )  
neglect it  $\rightarrow p_{id} = R$  (max of  $i_d$ )!

than current decreases for  $t > t_1$ ...  
(FORCING term negative)

so now we have a variable current

IDEA: choose  $t_2$   
(when switch SECTOR)  
such that after  $60^\circ$   
(sector period) the

current is still positive  $i_d \geq 0$



In this  $60^\circ$

I choose a suitable  $t_2$

to have at the end of  
this  $60^\circ$  an  $i_d \geq 0$  (NOT  $i_d < 0$ , corresponding to  $T_e < 0 \rightarrow \omega_m \downarrow$ )

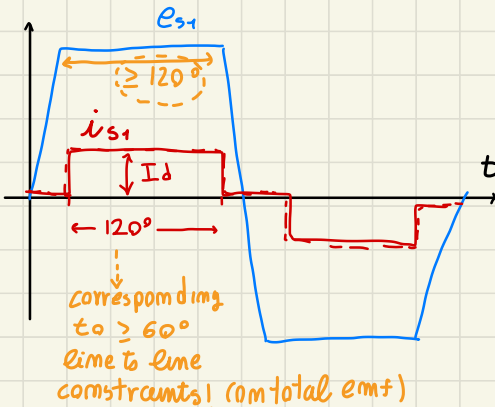
↓ after the  $60^\circ$  starting from  $t_2$  you have another sector (combination)  
[you need to anticipate the starting point of the sector  $t_2$ ,  
to have positive forcing term!]

looking to  $e_{mf}$  of 1<sup>st</sup> phase ( $e_{s1}$ )

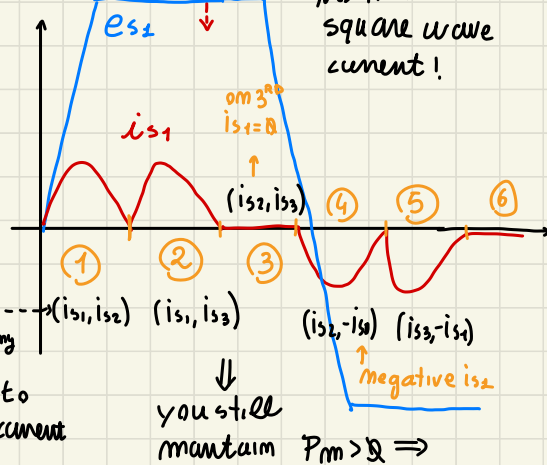
$\Rightarrow$

$\omega_m > \omega_b$  - strange current behaviour

$\omega_m < \omega_b$  ideal condition of work

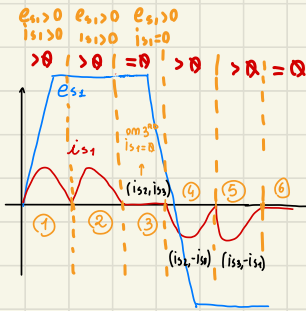


NO more  
square wave  
current!



mechanical power: (STILL POSITIVE)

$$P_m = e_{s1} i_{s1} + e_{s2} i_{s2} + e_{s3} i_{s3} > 0 \quad \text{mech power with oscillation (positive average } P_m)$$



same for other

phase  $e_s \cdot i_s \Rightarrow P_m > 0$  always with some oscillation (NOT fully constant  $P_m$ )

$\hookrightarrow$  positive average  $T_{e\text{avg}} > 0$ ,  $P_{m\text{avg}} > 0$

FOR  $\Omega_m > \Omega_b$

Let's check on the **DRAWBACKS** of this control idea:

the main DRAWBACK is not on small  $T_e$  oscillation, present in many machine! ( $T_e$  oscillation ok)

$\hookrightarrow$  BUT PROBLEM! looking to FIRST harmonic of current and emf

$\downarrow$  with an approach similar to AC

Brushless:

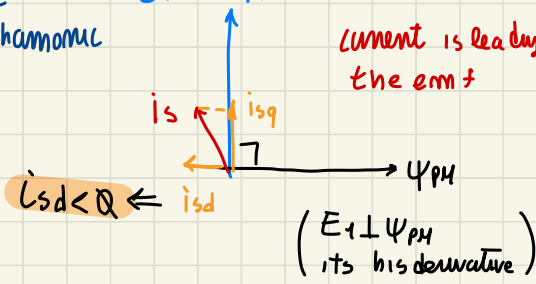
$\left\{ \begin{array}{l} 1^{\text{st}} \text{ current harmonic} \\ \text{is leading } 1^{\text{st}} \text{ emf harmonic} \end{array} \right.$

1<sup>st</sup> HARMONIC  $\downarrow$

In space phasor domain

$$E_1 = j\omega \psi_{PH}$$

current is leading the emf



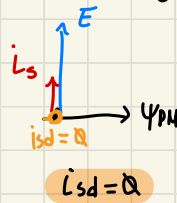
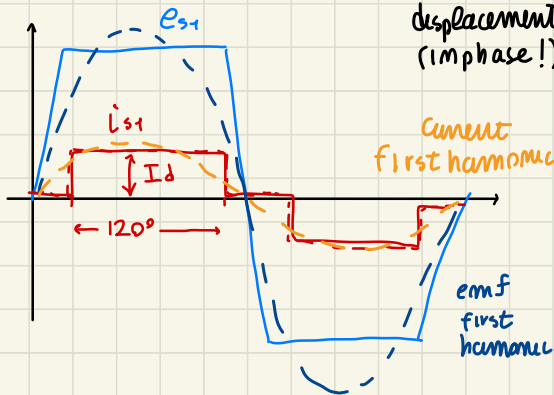
$$i_{sd} < 0 \leftarrow i_{sd}$$

( $E_1 \perp \psi_{PH}$  its derivative)

While for  $\Omega_m < \Omega_b$

→ they have same displacement (in phase!)

→ space phasor domain of 1 harmonic:



⇒ So, IF the 1 harmonic of current is leading emf, there is a demagnetizing ( $i_{sd} < 0$ ) current component  
↓ opposite to PM

STRATEGY with side effect... moving the waveform it demagnetize the PM

**BUT, because that's a LOW COST machine:**

⇒ PM are not good... so PM not able to demagnetize too much, or you reduce residual flux density (if demagnetized magnet it works Bad!)

↳ introduce this control strategy on DC Brushless if you have good PM on the machine (like Rare earth PM material)

↓ usually

it is better to go a little above  $\Omega_b$ ... (limit usage of this strategy)

$\Omega_m \sim 20\%$  above  $\Omega_b$  max! otherwise you have problem on PM behaviour

**NOTICE...**

for drone/elect scooter you reach  $\Omega_b$ , never  $\Omega_m > \Omega_b$ ... ← never demagnetizing approach  
(PROBLEMS in knowing sector of ROTOR...) → NOT aligned

you need costly sensors (encoders, ecc..)

⇒ problems on measurements (higher cost) of the instant, where is the ROTOR

with hall effect sensor...

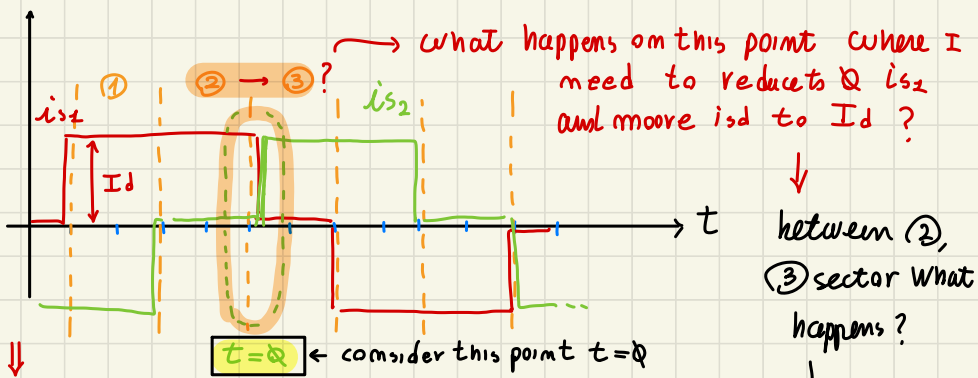
← this switching sector point IS HARD to recognize

you know the ideal switch point to from HALL EFFECT SENSOR!

↳ this machine has another problem! ⇒

Looking to equations on the equivalent circuit... **another machine PROBLEM:**

**PROBLEM** typical of square wave current electrical drive...



**② → ③** problems on switching

So when switching for small  $\Delta t$  you maintain  $i_{s1} > I_d$  decreasing while  $i_{s2} \uparrow > I_d$  and  $i_{s3} < I_d \rightarrow$  small period in which

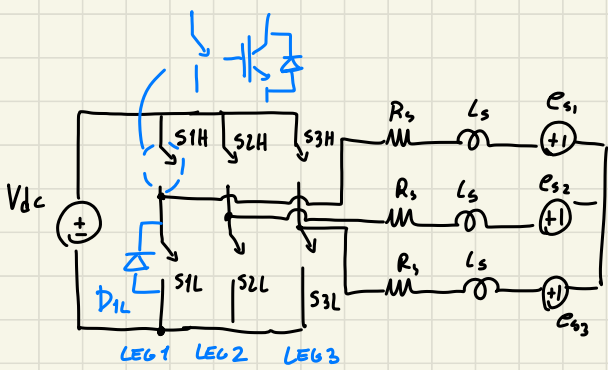
**THREE PHASE ON** (NOT 2 like on classical condition)

from  $t = 0$  on switching (2 → 3) point

$$\begin{cases} e_{s1}(0) = K_e \Omega_m \\ e_{s2}(0) = K_e \Omega_m \\ e_{s3}(0) = -K_e \Omega_m \end{cases} \quad \begin{array}{l} \text{(**) used later on} \\ \leftarrow \text{some overlapping because emf constant for at least } 120^\circ \text{ (simplification)} \end{array}$$

$$\begin{cases} i_{s1}(0^+) = I_d \\ i_{s2}(0^+) = 0 \\ i_{s3}(0^+) = -I_d \end{cases} \quad \begin{array}{l} \leftarrow \text{initial condition of transient} \\ (i_s(0^-) = i_s(0^+) \text{ because state variables}) \end{array}$$

• looking at **POWER CONVERTER**



(switch between sector ② → ③)

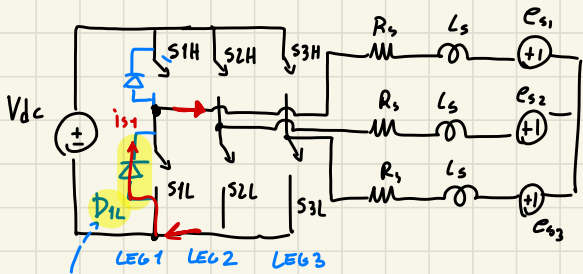
$$\begin{cases} t = 0^- \text{ leg 1 and leg 3} \\ t = 0^+ \text{ leg 2 and leg 3} \end{cases}$$



When  $m t = 0^+$  I goto the condition on which I open LEG1 completely:

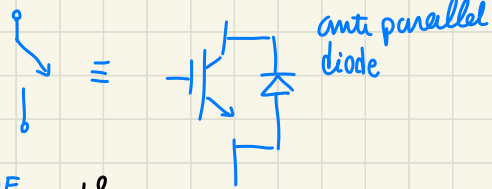
open both switch of FIRST LEG1

$(S1H=0, S1L=0) \rightarrow$  open LEG1



$$i_{S1}(0^+) = I_d \text{ (positive)}$$

remember that each switch  $s$  correspond to IGBT switch



When both open, there is also the DIODE... if

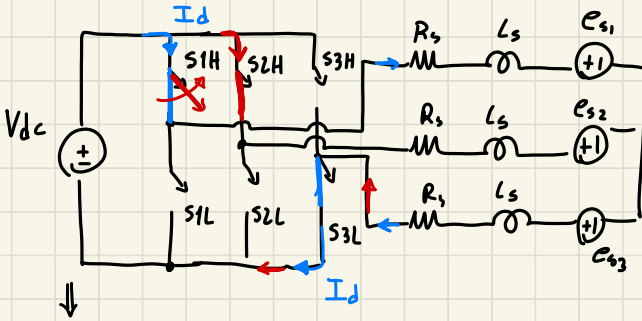
$i_{S2} = I_d \Rightarrow D1L$  start working! (low switches of LEG2)  
current continue flowing on that free wheeling diode!

considering:  
working switching

$$\begin{cases} t = 0^- \text{ leg 1 and leg 3} \\ t = 0^+ \text{ leg 2 and leg 3} \end{cases}$$

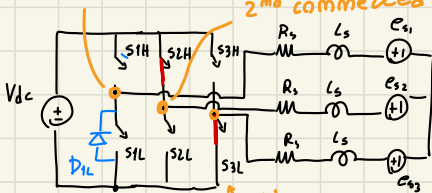
@  $t = 0^-$

@  $t = 0^+$



1st winding to neg. value of Vdc BUS

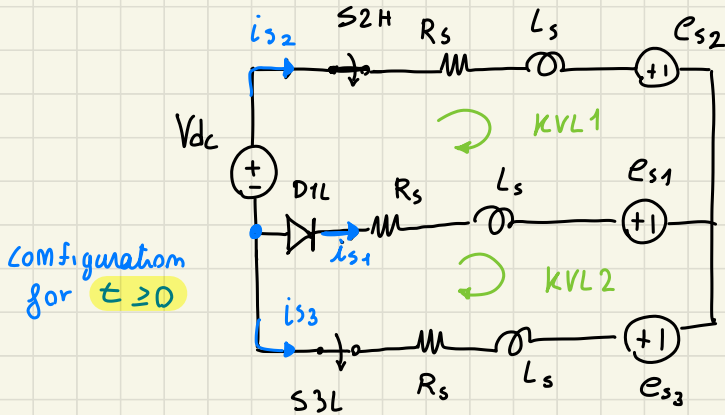
2nd connected to positive Vdc



3rd connected to neg. value of Vdc BUS

@  $t = 0^+$   
 $\Rightarrow$  equivalent circuit  $\Rightarrow$

⇒ With that proper connections equivalent circuit:



$$(KCL) \quad i_{s1} + i_{s2} + i_{s3} = 0 \Rightarrow i_{s3} = -i_{s1} - i_{s2}$$

$$\begin{cases} (KVL 1) & V_{dc} - R_s i_{s2} - L_s \frac{di_{s2}}{dt} - e_{s2} + e_{s1} + L_s \frac{di_{s1}}{dt} + R_s i_{s1} = 0 \\ (KVL 2) & -R_s i_{s1} - L_s \frac{di_{s3}}{dt} - e_{s1} + e_{s3} + L_s \frac{di_{s3}}{dt} + R_s i_{s3} = 0 \end{cases}$$

↑

neglecting voltage drop on D1L: negligible for low power applications...

↪ replacing  $i_{s3}$  on KVL 2 from KCL →

KVL 2... using  $i_{s3} = -i_{s1} - i_{s2}$

$$-R_s i_{s1} - L_s \frac{di_{s1}}{dt} - e_{s1} + e_{s3} - L_s \frac{di_{s1}}{dt} - L_s \frac{di_{s2}}{dt} - R_s i_{s1} - R_s i_{s2} = 0$$

↓

from this I can found  $i_{s2}$  as function of  $i_{s1}$

$$R_s i_{s2} + L_s \frac{di_{s2}}{dt} = -2R_s i_{s1} - 2L_s \frac{di_{s1}}{dt} - e_{s1} + e_{s3}$$

and then substitute  $i_{s2}$  on KVL 2, to have all in function of  $i_{s1}$

FROM (KVL 1)  $V_{dc} + 2R_s i_{s1} + 2L_s \frac{di_{s1}}{dt} + e_{s1} - e_{s3} - e_{s2} + e_{s1} + L_s \frac{di_{s1}}{dt} + R_s i_{s1} = 0$

↓ compute...

$$3R_s i_{s1} + 3L_s \frac{di_{s1}}{dt} = -V_{dc} \underbrace{(-e_{s1} + e_{s3} + e_{s2} - e_{s1})}_{-2K_e \Omega_m}$$

$$-K_e \Omega_m - K_e \Omega_m + K_e \Omega_m - K_e \Omega_m = -2K_e \Omega_m \quad (**)$$

↓

$$R_s i_{s1} + L_s \frac{di_{s1}}{dt} = \underbrace{\frac{-V_{dc} - 2K_e \Omega_m}{3}}_{< 0 \text{ (exponential decreasing)}} \quad (\text{to find } i_{s1})$$

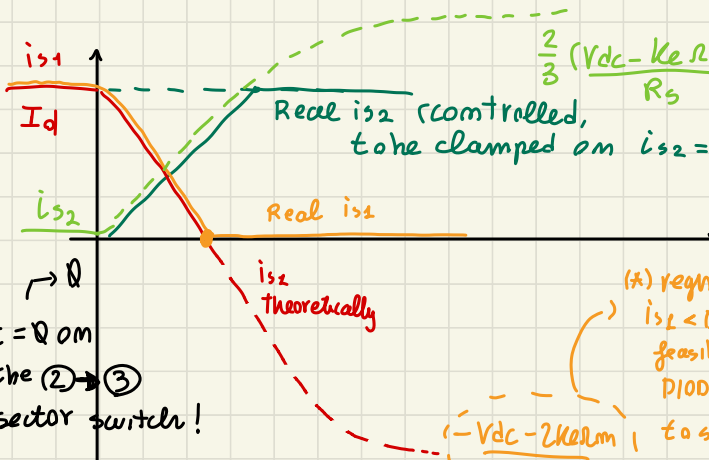
$$\begin{cases} e_{s1}(0) = K_e \Omega_m \\ e_{s2}(0) = K_e \Omega_m \\ e_{s3}(0) = -K_e \Omega_m \end{cases}$$

1<sup>ST</sup> ORDER transient with negative forcing term

↓  $i_{s1}(0^-) = i_{s1}(0^+) = I_d$ , then for  $t \geq 0$  it start transient

going exponentially to  $\frac{-V_{dc} - 2K_e \Omega_m}{3R_s}$  (\*) steady state with  $\tau = L_s/R_s \rightarrow$  reach steady state after  $5\tau$

theoretical  $i_{s2}(t)$



[ transient on  $i_{s2}$  has same  $\tau = L_s/R_s$ , BUT with different steady state conditions! ]

(\*) regime steady state condition of  $i_{s2} \dots$

$i_{s2} < 0 \dots$  BUT negative current not feasible because it go througher D1L DIODE, meaning that current has to stop on  $i_{s2} = 0$ , NO way to have  $i_{s1} < 0$  with both  $s1$  switches OPEN

$$\frac{-V_{dc} - 2K_e \Omega_m}{3R_s} \quad (5\tau)$$

them to find  $i_{s2}$  effect...

(FROM KVL 1)

$$\begin{aligned} e_{s2}(0) &= k_e \Omega_m \\ e_{s2}(0) &= k_e \Omega_m \end{aligned}$$

substitute from  $i_{s1}$  dym equation (\*)

$$V_{dc} - R_s i_{s2} - L_s \frac{di_{s2}}{dt} - \cancel{e_{s2}} + \cancel{e_{s1}} + (L_s \frac{di_{s1}}{dt} + R_s i_{s1}) = 0$$

$$R_s i_{s2} + L_s \frac{di_{s2}}{dt} = V_{dc} + \frac{-V_{dc} - 2k_e \Omega_m}{3} = \frac{2}{3} V_{dc} - \frac{2}{3} k_e \Omega_m$$

forcing term of IORD dynamic on  $i_{s2}$ ,

$\leftarrow (\geq 0?)$

from  $\Omega_0$  expression..

(NEGLECTING voltage drop on  $R_s$ )

$$\Omega_0 = \frac{V_{dc}}{2k_e} \quad \begin{matrix} \uparrow \\ \text{NO LOAD} \\ \text{speed } (\Omega_{max}) \end{matrix}$$

$$\Rightarrow @ \Omega_m = \Omega_0$$

substitute  $\Omega_0 = \frac{V_{dc}}{2k_e}$  on dym eq of  $i_{s2}$ :

$$R_s i_{s2} + L_s \frac{di_{s2}}{dt} = \frac{2}{3} V_{dc} - \frac{1}{3} V_{dc} = \frac{1}{3} V_{dc} > 0$$

also on  $\Omega_m = \Omega_0$  NO LOAD speed we have a positive forcing term

$\leftarrow$  so we can represent

$i_{s2}(t)$  on previous

plot... increasing with t

reaching ideally

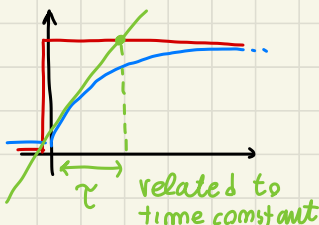
$$\frac{(2/3 V_{dc} - 2/3 k_e \Omega_m)}{R_s}$$

(forcing term always  $> 0$ ) until  $\Omega_0$  (OR  $\Omega_b \sim \Omega_0$ )

( $i_{s1}, i_{s2}$  has same  $\tau$  but different steady state condition)

$i_{s1}, i_{s2}$  slopes depends on steady state conditions...

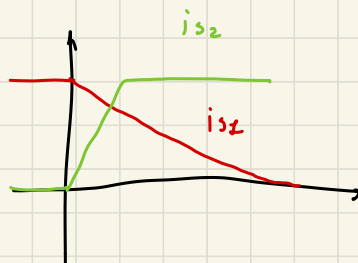
Remember, In IORD transient

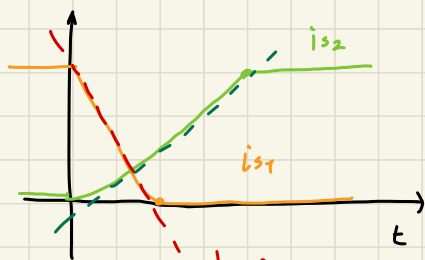


$\Rightarrow$  so given same  $\tau = L_s/R_s$  for both  $i_{s1}, i_{s2}$ , BUT different final value  $\Rightarrow$  you have different slope of  $i_s(t)$  function

$\hookrightarrow$  you may have: 2 different slopes

$\leftarrow$  (condition different from previous plot)



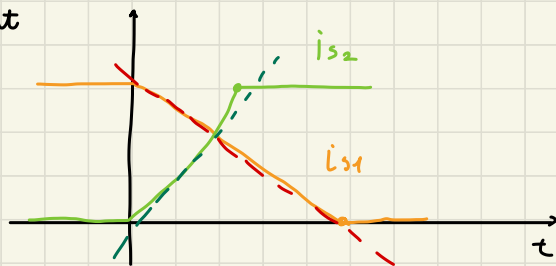


higher slope on  $i_{s2}$ :

meaning that

$$\left| \frac{-V_{dc} - 2k_e \Omega_m}{3} \right| > \frac{2(V_{dc} - k_e \Omega_m)}{3 R_s}$$

different slopes



while on this different condition:

$$\left| \frac{-V_{dc} - 2k_e \Omega_m}{3} \right| < \frac{2(V_{dc} - k_e \Omega_m)}{3 R_s}$$

because  $i_{s2}$  has higher slope!

$\tau$  depends on: comparing the steady state conditions  
of  $i_{s1}$ ,  $i_{s2}$

$$\underbrace{\frac{2(V_{dc} - k_e \Omega_m)}{3 R_s}}_{i_{s2}(\infty)} = \underbrace{\frac{V_{dc} + 2k_e \Omega_m}{3 R_s}}_{i_{s1}(\infty)}$$

$$V_{dc} = 4 k_e \Omega_m \Rightarrow \hat{\Omega}_m = \frac{V_{dc}}{4 k_e} = \frac{1}{2} \Omega_o$$

so:

$$\text{IF } \Omega_m > \hat{\Omega}_m \uparrow$$

$$\left. \begin{array}{l} \frac{2(V_{dc} - k_e \Omega_m)}{3 R_s} \downarrow \downarrow \text{decrease } (i_{s2}(\infty)) \\ \frac{V_{dc} + 2k_e \Omega_m}{3 R_s} \uparrow \uparrow \text{increase } (i_{s1}(\infty)) \end{array} \right\}$$

meaning that  $i_{s2}$  has an higher slope than  $i_{s1}$

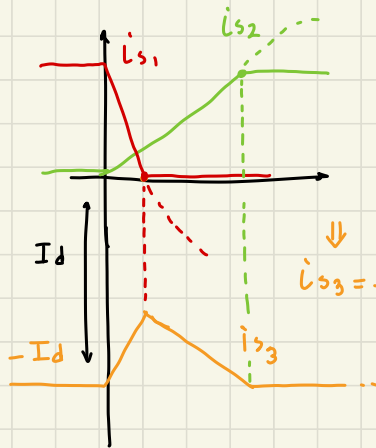
viceversa, for  $\Omega_m < \hat{\Omega}_m \downarrow \Rightarrow i_{s2}$  has higher slope than  $i_{s1}$

$\Rightarrow \dots$

speed at which they have same slope, same regime value



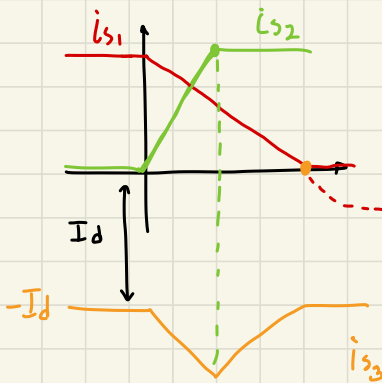
- for **high speed**: ( $i_{s2}$  has higher slope than  $i_{s1}$ )



than looking to  $i_{s3}$ , NOT constant

$$i_{s3} = -i_{s1} - i_{s2}$$

- for **low speed** ( $i_{s2}$  has higher speed than  $i_{s1}$ )



$$i_{s3} = -i_{s1} - i_{s2}$$

There is a change in  $i_{s3}$ !

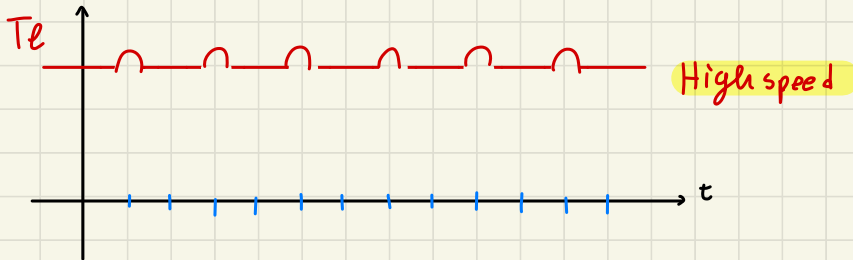
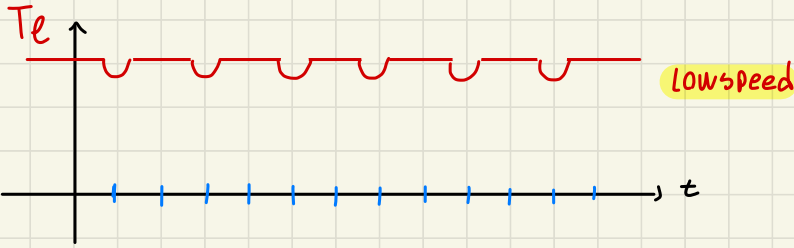
↓  
Oscillation in  $i_{s3}$ , with  
different approach if you are  
in HIGH speed OR LOW SPEED...

↓  
this oscillation in  $i_{s3}$  means  
that mech power during  
Three phase on period  
is NOT constant (not constant  $I_d$   
but some combinations, current  
oscillation) ⇒

In this machine **PROBLEM:**



on each sector: **NOT overall constant torque!**



(NOT constant Torque)!



When you change the SECTOR you have some noise!

(noisy machine)

due to the fact that you cannot instantaneously reduce to 0 the current when switching!

(no 0 current on coil instantaneously)



Transient is →  $P_m$  not constant!



when change sector  $T_e$  oscillate!

torque oscillation when change sector



With some control strategy

you can handle this commutation  $T_e$ ...

(control properly the current to face sector commutation)



next step: switched reluctance machine: